

Worked Solutions

1. Calculus, 2ADV C3 2022 HSC 22

$$y = x^3 - 6x^2 + 8$$

$$\frac{dy}{dx} = 3x^2 - 12x$$

$$\frac{d^2y}{dx^2} = 6x - 12$$

$$\text{SP's when } \frac{dy}{dx} = 0:$$

$$3x^2 - 12x = 0$$

$$3x(x - 4) = 0$$

$$x = 0 \text{ or } 4$$

$$\text{When } x = 0, y = 8, \frac{d^2y}{dx^2} < 0$$

$$\Rightarrow \text{Local Max at } (0, 8)$$

$$\text{When } x = 4, y = 4^3 - 6(4^2) + 8 = -24, \frac{d^2y}{dx^2} > 0$$

$$\Rightarrow \text{Local Min at } (4, -24)$$

Check ends of domain:

$$\text{When } x = -1, y = -1 - 6 + 8 = 1$$

$$\text{When } x = 7, y = 7^3 - 6(7^2) + 8 = 57$$

$$\therefore \text{Global max} = 57$$

$$\therefore \text{Global min} = -24$$

Worked Solutions

2. Calculus, 2ADV C3 2004 HSC 9c

i. $f(x) = \frac{\log_e x}{x}$, for $x > 0$

Using the quotient rule,

$$f'(x) = \frac{u'v - uv'}{v^2}$$

$$\begin{aligned} f'(x) &= \frac{x \cdot \frac{d}{dx}(\log_e x) - \log_e x \cdot \frac{d}{dx}(x)}{x^2} \\ &= \frac{x\left(\frac{1}{x}\right) - \log_e x \cdot 1}{x^2} \\ &= \frac{1 - \log_e x}{x^2} \end{aligned}$$

$$\text{SP's occur when } f'(x) = 0$$

$$\frac{1 - \log_e x}{x^2} = 0$$

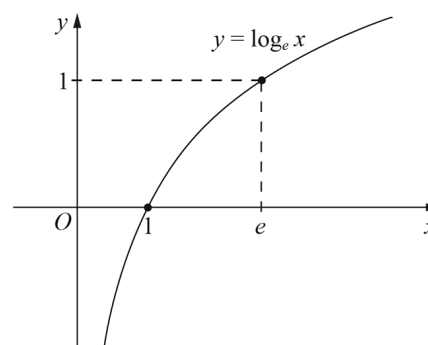
$$1 - \log_e x = 0$$

$$\log_e x = 1$$

$$\therefore x = e$$

$\therefore$ There is a stationary point at $x = e$.

ii.



When $x < e$, $\log_e x < 1$ (from graph)

$$1 - \log_e x > 0$$

$$\frac{1 - \log_e x}{x^2} > 0$$

$$\therefore f'(x) > 0$$

When $x > e$, $\log_e x > 1$ (from graph)

$$1 - \log_e x < 0$$

$$< 0$$

$$\frac{1 - \log_e x}{x^2}$$

$$\therefore f'(x) < 0$$

$\therefore$ The stationary point at $x = e$ is a maximum.

iii. $f(e) = \frac{\log_e x}{e}$

$$= \frac{1}{e}$$

$\therefore f(x)$ has a maximum value at $\left(e, \frac{1}{e}\right)$

$$\therefore \frac{\log_e x}{x} \leq \frac{1}{e}$$

$$\log_e x \leq \frac{x}{e}$$

$$e \log_e x \leq x$$

$$\log_e x^e \leq x$$

$$e^{\log_e x^e} \leq e^x$$

$$x^e \leq e^x \quad (\text{using } e^{\log x} = x)$$

$$\therefore e^x \geq x^e \quad (\text{for } x > 0)$$

MARKER'S COMMENT: Very challenging for most students. Successful students recognised the link to part (ii).

3. Calculus, 2ADV C3 2009 HSC 10

a. $f(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$

Turning points when $f'(x) = 0$

$$f'(x) = 1 - x + x^2$$

$$x^2 - x + 1 = 0$$

$$\text{Since } \Delta = b^2 - 4ac$$

$$= (-1)^2 - 4 \times 1 \times 1$$

$$= -3 < 0 \Rightarrow \text{No solution}$$

$\therefore f(x)$ has no turning points

b. P.I. when $f''(x) = 0$

$$f''(x) = -1 + 2x = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

Check for change in concavity

$$f''\left(\frac{1}{4}\right) = -\frac{1}{2} < 0$$

$$f''\left(\frac{3}{4}\right) = \frac{1}{2} > 0$$

$\Rightarrow$ Change in concavity

$$\therefore \text{P.I. at } x = \frac{1}{2}$$

$$f\left(\frac{1}{2}\right) = \frac{1}{2} - \frac{\left(\frac{1}{2}\right)^2}{2} + \frac{\left(\frac{1}{2}\right)^3}{3}$$

$$= \frac{1}{2} - \frac{1}{8} + \frac{1}{24}$$

$$= \frac{5}{12}$$

$\therefore$ Point of Inflection at $\left(\frac{1}{2}, \frac{5}{12}\right)$

c.i. Show $1 - x + x^2 - \frac{1}{1+x} = \frac{x^3}{1+x}, \quad x \neq -1$

$$\text{LHS} = \frac{1+x}{1+x} - \frac{x(1+x)}{1+x} + \frac{x^2(1+x)}{(1+x)} - \frac{1}{1+x}$$

$$= \frac{1+x - x - x^2 + x^2 + x^3 - 1}{1+x}$$

♦♦ Mean mark 28% for all of Q10
(note that data for each question part is not available).

$$= \frac{x^3}{1+x} \quad \dots \text{ as required}$$

c.ii. Let $g(x) = \ln(1+x)$

$$g'(x) = \frac{1}{1+x}$$

$$f'(x) - g'(x) = 1 - x + x^2 - \frac{1}{1+x}$$

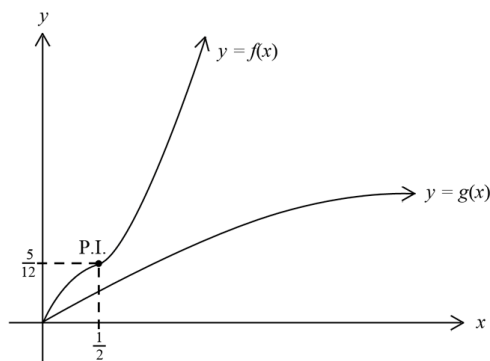
$$= \frac{x^3}{1+x} \quad (\text{using part (i)})$$

Since $\frac{x^3}{1+x} \geq 0$ for $x \geq 0$

$$f'(x) - g'(x) \geq 0$$

$$f'(x) \geq g'(x) \text{ for } x \geq 0$$

d.



e. Show $\frac{d}{dx} [(1+x)\ln(1+x) - (1+x)] = \ln(1+x)$

Using $\frac{d}{dx} uv = uv' + vu'$

$$\text{LHS} = (1+x) \times \frac{1}{1+x} + \ln(1+x) \times 1 + -1$$

$$= 1 + \ln(1+x) - 1$$

$$= \ln(1+x)$$

$$= \text{RHS} \quad \dots \text{ as required}$$

f. Area = $\int_0^1 f(x) - g(x) \, dx$

$$= \int_0^1 \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \ln(x+1) \right) dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{12} - (1+x)\ln(1+x) + (1+x) \right]_0^1$$

(using part (e) above)

$$= \left[\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{12} - (2)\ln 2 + 2 \right) - (\ln 1 + 1) \right]$$

$$= \frac{5}{12} - 2\ln 2 + 2 - 1$$

$$= 1 \frac{5}{12} - 2\ln 2 \quad \text{u}^2$$

MARKER'S COMMENT: When 2 graphs are drawn on the same set of axes, you **must** label them.

4. Calculus, 2ADV C3 2012 HSC 14a

i. $f(x) = 3x^4 + 4x^3 - 12x^2$
 $f'(x) = 12x^3 + 12x^2 - 24x$
 $f''(x) = 36x^2 + 24x - 24$

Stationary points when $f'(x) = 0$

$$12x^3 + 12x^2 - 24x = 0$$

$$12x(x^2 + x - 2) = 0$$

$$12x(x + 2)(x - 1) = 0$$

$\therefore$ Stationary points at $x = 0, 1$ or -2

$$\text{When } x = 0, \quad f(0) = 0$$

$$f''(0) = -24 < 0$$

$\therefore$ MAX at $(0, 0)$

When $x = 1$

$$f(1) = 3 + 4 - 12 = -5$$

$$f''(1) = 36 + 24 - 24 = 36 > 0$$

$\therefore$ MIN at $(1, -5)$

When $x = -2$

$$f(-2) = 3(-2)^4 + 4(-2)^3 - 12(-2)^2$$

$$= 48 - 32 - 48$$

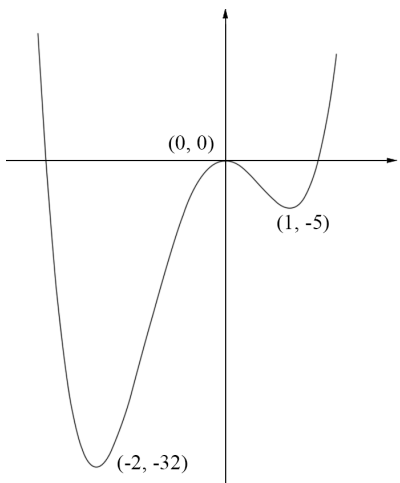
$$= -32$$

$$f''(-2) = 36(-2)^2 + 24(-2) - 24$$

$$= 144 - 48 - 24 = 72 > 0$$

$\therefore$ MIN at $(-2, -32)$

ii.



iii. $f(x)$ is increasing for
 $-2 < x < 0$ and $x > 1$

iv. Find k such that
 $3x^4 + 4x^3 - 12x^2 + k = 0$ has no solution
 k is the vertical shift of $y = 3x^4 + 4x^3 - 12x^2$
 $\Rightarrow$ No solution if it does not cross the x -axis.
 $\therefore$ No solution when $k > 32$

◆ Mean mark 42%
MARKER'S COMMENT: Be careful to use the correct inequality signs, and not carelessly include $\geq$ or $\leq$ by mistake.

◆◆ Mean mark 12%.